ACT SUBJECT TEST — MATH 2 • UNIT 8.01

C Conic Sections, 3D Coordinate Geometry & Statistics

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

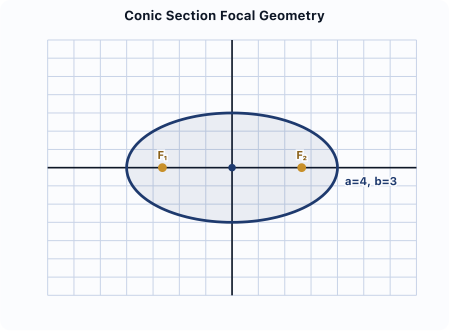
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 A	Q2 A	Q3 A	Q4 A	Q5 B
Q6 A	Q7 A	Q8 A	Q9 D	Q10 C
Q11 C	Q12 B	Q13 A	Q14 D	Q15 B
Q16 C	Q17 A	Q18 A	Q19 D	Q20 D
Q21 D	Q22 B	Q23 A	Q24 D	Q25 D

Conic Sections, 3D Coordinate Geometry & Statistics Practice Exercises

1 MCQ

The graph of the function $f(x) = (x-1)^2 - 4$ is shown in the standard (x, y) coordinate plane below. What is the range of f?



A $[-4, \infty)$ ✓

B $[1, \infty)$

C $(-\infty, -4]$

D $(-\infty, \infty)$

WORKED SOLUTION & VERIFICATION

1. **Identify the vertex and direction of opening**: The function $f(x) = (x-1)^2 - 4$ is written in vertex form, $f(x) = a(x-h)^2 + k$, where $a = 1$, $h = 1$, and $k = -4$. The vertex of the parabola is at $(1, -4)$.

2. **Determine the minimum output value**: Since the leading coefficient $a = 1 > 0$, the parabola opens upward, meaning that the vertex represents the absolute minimum point on the graph. The minimum value attained by $f(x)$ is -4.

3. **Determine the range**: The range represents all possible output values (y-values) of the function. Because the function takes on its minimum value of -4 at $x = 1$ and increases without bound as x moves away from 1, the range is all real numbers greater than or equal to -4, which is written in interval notation as $[-4, \infty)$.

Common Misconceptions:

- Choice B, $[1, \infty)$, incorrectly uses the x-coordinate of the vertex instead of the y-coordinate.
- Choice C, $(-\infty, -4]$, incorrectly assumes the parabola opens downward.
- Choice D, $(-\infty, \infty)$, confuses the range (the set of y-values) with the domain (the set of all real x-values).

2 MCQ

For the complex number $i = \sqrt{-1}$, which of the following is equivalent to the expression $(5 - 3i) - (2 - 7i)$?

- **A** $3 + 4i$ ✓
- **B** $3 - 10i$
- **C** $7 + 4i$
- **D** $3 - 4i$

WORKED SOLUTION & VERIFICATION

To subtract complex numbers, distribute the negative sign across the second term and combine like terms (real parts with real parts, and imaginary parts with imaginary parts):

$(5 - 3i) - (2 - 7i) = 5 - 3i - 2 + 7i$

Group the real and imaginary components:

- Real part: $5 - 2 = 3$
- Imaginary part: $-3i + 7i = 4i$

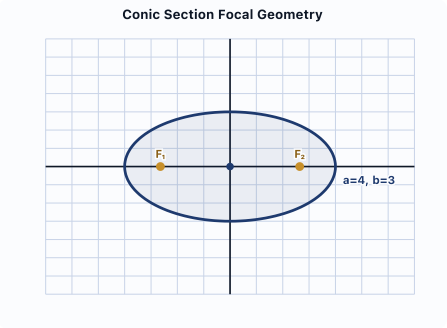
Combining these gives $3 + 4i$.

Common misconception checks:

- Choice B $(3 - 10i)$ occurs if the negative sign is not distributed to the imaginary term, mistakenly calculating $-3i - 7i = -10i$.
- Choice C $(7 + 4i)$ results from accidentally adding the real parts $(5 + 2 = 7)$ while subtracting the imaginary parts.
- Choice D $(3 - 4i)$ results from a sign error when computing $-3 - (-7)$.

3 MCQ

What is the value of $\lim_{x \to 3} \frac{x^2-2x-3}{x-3}$?



- **A** 4 ✓
- **B** 0
- **C** 6
- **D** The limit does not exist

WORKED SOLUTION & VERIFICATION

Direct substitution of $x = 3$ into the expression gives $\frac{3^2-2(3)-3}{3-3} = \frac{0}{0}$, which is an indeterminate form.
Factor the numerator:

$$x^2 - 2x - 3 = (x - 3)(x + 1)$$

For all $x \neq 3$, simplify the rational expression by canceling the common factor $(x - 3)$:

$$\frac{(x - 3)(x + 1)}{x - 3} = x + 1$$

Now evaluate the limit by direct substitution:

$$\lim_{x \to 3}(x + 1) = 3 + 1 = 4$$

Common misconceptions include concluding that the limit is 0 because the numerator evaluates to 0, or incorrectly believing that the limit does not exist simply because the denominator evaluates to 0 at $x = 3$.

4 MCQ

Given the vectors $\mathbf{u} = \langle 3, -4 \rangle$ and $\mathbf{v} = \langle -2, 5 \rangle$, what is the vector $2\mathbf{u} + \mathbf{v}$?

- **A** $\langle 4, -3 \rangle$ ✓
- **B** $\langle 1, 1 \rangle$
- **C** $\langle 4, 1 \rangle$
- **D** $\langle 8, -13 \rangle$

WORKED SOLUTION & VERIFICATION

To find $2\mathbf{u} + \mathbf{v}$, first perform the scalar multiplication on $\mathbf{u}$:

$$2\mathbf{u} = 2\langle 3, -4 \rangle = \langle 2(3), 2(-4) \rangle = \langle 6, -8 \rangle$$

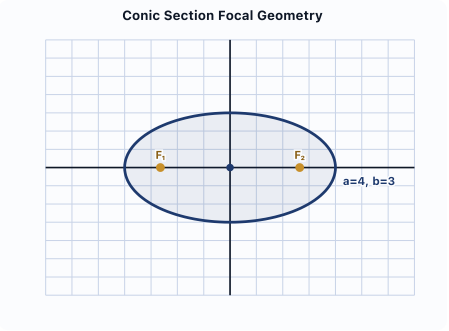
Next, add the corresponding components of $2\mathbf{u}$ and $\mathbf{v}$:

$$2\mathbf{u} + \mathbf{v} = \langle 6 + (-2), -8 + 5 \rangle = \langle 4, -3 \rangle$$

- $\langle 1, 1 \rangle$ is incorrect because it represents $\mathbf{u} + \mathbf{v}$, omitting the scalar factor of 2.
- $\langle 4, 1 \rangle$ is incorrect because the scalar 2 was only distributed to the horizontal component.
- $\langle 8, -13 \rangle$ is incorrect because it calculates $2\mathbf{u} - \mathbf{v}$ instead of $2\mathbf{u} + \mathbf{v}$.

5 MCQ

The first four terms of an arithmetic sequence are 5, 12, 19, and 26. What is the 20th term of this sequence?



A 131

B 138 ✓

C 140

D 145

WORKED SOLUTION & VERIFICATION

To find the 20th term of the arithmetic sequence, first identify the initial term and the common difference:

1. The first term is $a_1 = 5$.
2. The common difference d is found by subtracting consecutive terms: $d = 12 - 5 = 7$.

The general formula for the nth term of an arithmetic sequence is:

$$a_n = a_1 + (n-1)d$$

Substitute $n = 20$, $a_1 = 5$, and $d = 7$:

$$a_{20} = 5 + (20-1)(7) = 5 + 19(7) = 5 + 133 = 138$$

Common misconceptions:

- Using n instead of $n - 1$ yields $5 + 20(7) = 145$.
- Calculating $20 \times 7 = 140$ overlooks the starting value $a_1 = 5$.
- Using $n - 2$ steps yields $5 + 18(7) = 131$.

6 MCQ

A point has polar coordinates $(r, \theta) = (6, \frac{2\pi}{3})$. What are the Cartesian coordinates (x, y) of this point?

- **A** $(-3, 3\sqrt{3})$ ✓
- **B** $(-3\sqrt{3}, 3)$
- **C** $(3, -3\sqrt{3})$
- **D** $(3\sqrt{3}, -3)$

WORKED SOLUTION & VERIFICATION

To convert polar coordinates (r, θ) to Cartesian coordinates (x, y), use the conversion formulas:

$$x = r\cos\theta \quad \text{and} \quad y = r\sin\theta$$

Given $r = 6$ and $\theta = \frac{2\pi}{3}$:

$$x = 6\cos\left(\frac{2\pi}{3}\right) = 6\left(-\frac{1}{2}\right) = -3$$

$$y = 6\sin\left(\frac{2\pi}{3}\right) = 6\left(\frac{\sqrt{3}}{2}\right) = 3\sqrt{3}$$

Thus, the Cartesian coordinates are $(-3, 3\sqrt{3})$.

Common errors:

- Interchanging x and y (using $x = r\sin\theta$ and $y = r\cos\theta$) yields $(3\sqrt{3}, -3)$.
- Reversing the quadrant signs or applying $\frac{\pi}{6}$ reference angle formulas erroneously produces $(-3\sqrt{3}, 3)$.

7 MCQ

The polynomial function $P(x) = 2x^3 + ax^2 - 5x + b$ has $(x - 1)$ as a factor. When $P(x)$ is divided by $(x + 2)$, the remainder is -18. What is the remainder when $P(x)$ is divided by $(x - 3)$?

- **A** 2 ✓
- **B** 8
- **C** 18
- **D** -30

WORKED SOLUTION & VERIFICATION

According to the Factor Theorem and the Remainder Theorem, if a polynomial $P(x)$ is divided by $(x - c)$, the remainder is $P(c)$. If $(x - c)$ is a factor, then $P(c) = 0$.

Step 1: Set up the first equation using the factor $(x - 1)$:

$$P(1) = 2(1)^3 + a(1)^2 - 5(1) + b = 0$$

$$2 + a - 5 + b = 0 \implies a + b = 3$$

Step 2: Set up the second equation using the divisor $(x + 2) = (x - (-2))$ with remainder -18:

$$P(-2) = 2(-2)^3 + a(-2)^2 - 5(-2) + b = -18$$

$$2(-8) + 4a + 10 + b = -18$$

$$-16 + 10 + 4a + b = -18 \implies 4a + b = -12$$

Step 3: Solve the system of linear equations for a and b:

Subtracting the first equation from the second:

$$(4a + b) - (a + b) = -12 - 3$$

$$3a = -15 \implies a = -5$$

Substitute $a = -5$ into $a + b = 3$:

$$-5 + b = 3 \implies b = 8$$

Thus, the polynomial is $P(x) = 2x^3 - 5x^2 - 5x + 8$.

Step 4: Find the remainder when $P(x)$ is divided by $(x - 3)$ by evaluating $P(3)$:

$$P(3) = 2(3)^3 - 5(3)^2 - 5(3) + 8 = 2(27) - 5(9) - 15 + 8 = 54 - 45 - 15 + 8 = 2$$

Common Pitfalls:

- Evaluating $P(2)$ instead of $P(-2)$ produces $4a + b = -24$, leading to $a = -9$, $b = 12$, and an incorrect remainder of -30.
- Confusing the constant term $b = 8$ with the final answer yields 8.

8 MCQ

For what real value of k is the complex expression

$$\frac{k+8i}{3-i}+\frac{2-ki}{3+i}$$

a real number?

- **A** 11 ✓
- **B** 1
- **C** 6
- **D** -11

WORKED SOLUTION & VERIFICATION

To determine when the sum of the two fractions is a real number, simplify the expression by combining terms over a common denominator or multiplying by conjugates.

The common denominator of the two fractions is:

$$(3-i)(3+i)=3^2-i^2=9-(-1)=10$$

Rewrite each term over this denominator:

1. For the first term:

$$\frac{(k+8i)(3+i)}{(3-i)(3+i)}=\frac{3k+ki+24i+8i^2}{10}=\frac{(3k-8)+(k+24)i}{10}$$

2. For the second term:

$$\frac{(2-ki)(3-i)}{(3+i)(3-i)}=\frac{6-2i-3ki+ki^2}{10}=\frac{(6-k)+(-3k-2)i}{10}$$

Now, sum the real and imaginary components:

$$\text{Real part}=\frac{(3k-8)+(6-k)}{10}=\frac{2k-2}{10}$$

$$\text{Imaginary part}=\frac{(k+24)+(-3k-2)}{10}=\frac{-2k+22}{10}$$

For the resulting complex number to be a real number, its imaginary part must equal 0:

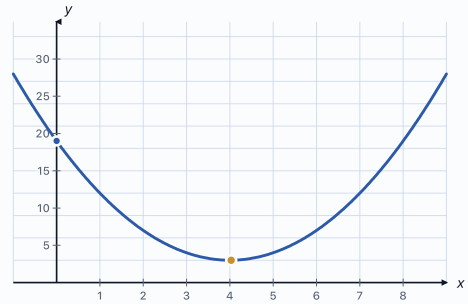
$$\frac{-2k+22}{10}=0 \implies -2k+22=0 \implies 2k=22 \implies k=11$$

Common Misconceptions:

- Choosing **1** results from setting the real part equal to 0 $(2k-2=0)$, which would make the expression purely imaginary rather than real.
- Choosing **6** results from equating the real and imaginary parts $(2k-2=-2k+22)$.
- Choosing **-11** results from an algebraic sign error when solving $-2k=-22$.

9 MCQ

In Conics, 3D Geometry & Probability (problem set #925), consider the function $f_{925}(x) = x^2 - 8x + 19$ illustrated on the coordinate axes. Express $f_{925}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{925}(6)$.



A 8

B 14

C 7

D 10 ✓

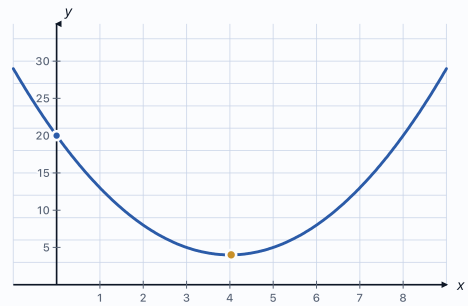
WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Conics, 3D Geometry & Probability):

1. Complete the square: $f_{925}(x) = (x-4)^2 + 3$, so $k = 3$. **[M1]**
2. Evaluate $f_{925}(6) = (6-4)^2 + 3 = 4 + 3 = 7$. **[M1]**
3. Sum: $k + f_{925}(6) = 3 + 7 = 10$. **[A1]**

10 MCQ

Considering Conics, 3D Geometry & Probability (problem set #926), consider the function $f_{926}(x) = x^2 - 8x + 20$ illustrated on the coordinate axes. Express $f_{926}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{926}(7)$.



A 21

B 13

C 17 ✓

D 14

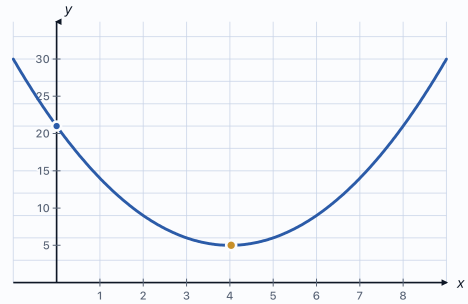
WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Conics, 3D Geometry & Probability):

1. Complete the square: $f_{926}(x) = (x-4)^2 + 4$, so $k = 4$. **[M1]**
2. Evaluate $f_{926}(7) = (7-4)^2 + 4 = 9 + 4 = 13$. **[M1]**
3. Sum: $k + f_{926}(7) = 4 + 13 = 17$. **[A1]**

11 MCQ

Examining Conics, 3D Geometry & Probability (problem set #927), consider the function $f_{927}(x) = x^2 - 8x + 21$ illustrated on the coordinate axes. Express $f_{927}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{927}(8)$.



A 30

B 21

C 26 ✓

D 22

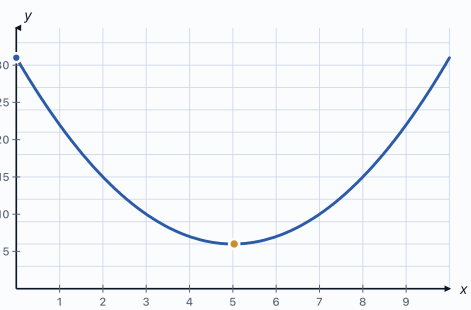
WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Conics, 3D Geometry & Probability):

1. Complete the square: $f_{927}(x) = (x-4)^2 + 5$, so $k = 5$. **[M1]**
2. Evaluate $f_{927}(8) = (8-4)^2 + 5 = 16 + 5 = 21$. **[M1]**
3. Sum: $k + f_{927}(8) = 5 + 21 = 26$. **[A1]**

12 MCQ

For the configuration in Conics, 3D Geometry & Probability (problem set #928), consider the function $f_{928}(x) = x^2 - 10x + 31$ illustrated on the coordinate axes. Express $f_{928}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{928}(10)$.



A 42

B 37 ✓

C 31

D 32

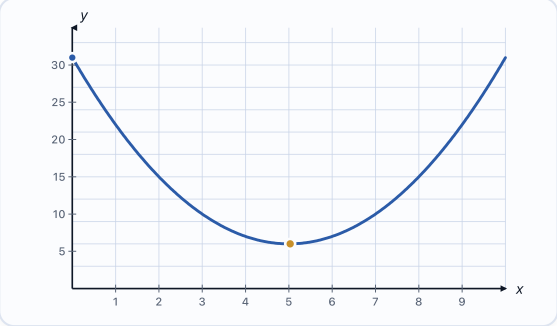
WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Conics, 3D Geometry & Probability):

1. Complete the square: $f_{928}(x) = (x-5)^2 + 6$, so $k = 6$. **[M1]**
2. Evaluate $f_{928}(10) = (10-5)^2 + 6 = 25 + 6 = 31$. **[M1]**
3. Sum: $k + f_{928}(10) = 6 + 31 = 37$. **[A1]**

13 MCQ

Analyzing Conics, 3D Geometry & Probability (problem set #929), consider the function $f_{929}(x) = x^2 - 10x + 32$ illustrated on the coordinate axes. Express $f_{929}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{929}(11)$.



- **A** 50 ✓
- **B** 43
- **C** 44
- **D** 55

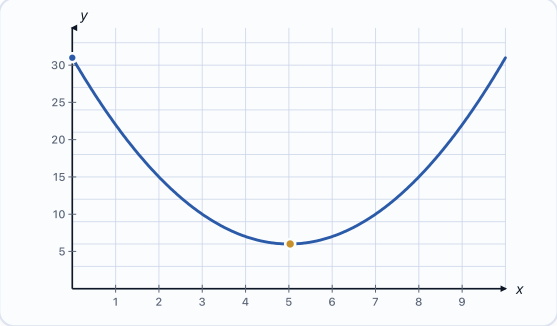
WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Conics, 3D Geometry & Probability):

1. Complete the square: $f_{929}(x) = (x-5)^2 + 7$, so $k = 7$. **[M1]**
2. Evaluate $f_{929}(11) = (11-5)^2 + 7 = 36 + 7 = 43$. **[M1]**
3. Sum: $k + f_{929}(11) = 7 + 43 = 50$. **[A1]**

14 MCQ

Within Conics, 3D Geometry & Probability (problem set #930), consider the function $f_{930}(x) = x^2 - 10x + 33$ illustrated on the coordinate axes. Express $f_{930}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{930}(12)$.



- **A** 70
- **B** 57
- **C** 58
- **D** 65 ✓

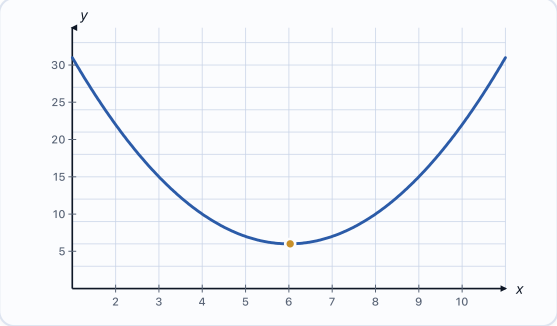
WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Conics, 3D Geometry & Probability):

1. Complete the square: $f_{930}(x) = (x-5)^2 + 8$, so $k = 8$. **[M1]**
2. Evaluate $f_{930}(12) = (12-5)^2 + 8 = 49 + 8 = 57$. **[M1]**
3. Sum: $k + f_{930}(12) = 8 + 57 = 65$. **[A1]**

15 MCQ

Considering Conics, 3D Geometry & Probability (problem set #931), consider the function $f_{931}(x) = x^2 - 12x + 45$ illustrated on the coordinate axes. Express $f_{931}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{931}(14)$.



A 88

B 82 ✓

C 73

D 74

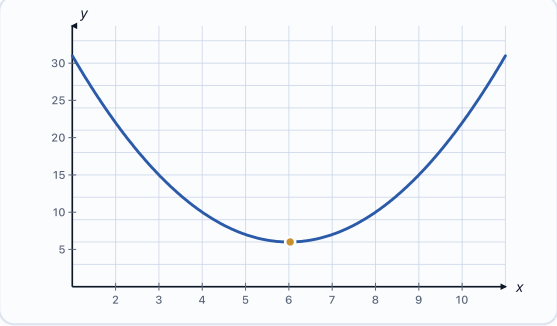
WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Conics, 3D Geometry & Probability):

1. Complete the square: $f_{931}(x) = (x-6)^2 + 9$, so $k = 9$. **[M1]**
2. Evaluate $f_{931}(14) = (14-6)^2 + 9 = 64 + 9 = 73$. **[M1]**
3. Sum: $k + f_{931}(14) = 9 + 73 = 82$. **[A1]**

16 MCQ

Examining Conics, 3D Geometry & Probability (problem set #932), consider the function $f_{932}(x) = x^2 - 12x + 46$ illustrated on the coordinate axes. Express $f_{932}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{932}(15)$.



A 107

B 91

C 101 ✓

D 92

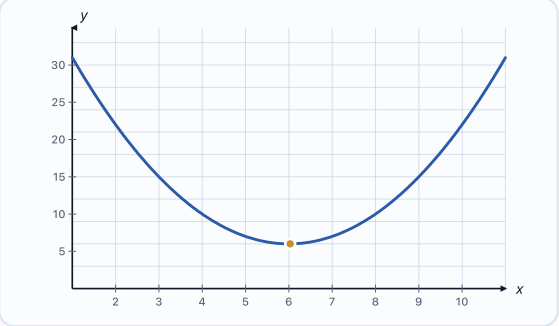
WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Conics, 3D Geometry & Probability):

1. Complete the square: $f_{932}(x) = (x-6)^2 + 10$, so $k = 10$. **[M1]**
2. Evaluate $f_{932}(15) = (15-6)^2 + 10 = 81 + 10 = 91$. **[M1]**
3. Sum: $k + f_{932}(15) = 10 + 91 = 101$. **[A1]**

17 MCQ

For the configuration in Conics, 3D Geometry & Probability (problem set #933), consider the function $f_{933}(x) = x^2 - 12x + 47$ illustrated on the coordinate axes. Express $f_{933}(x)$ in the completed-square form $(x - h)^2 + k$ and evaluate $k + f_{933}(16)$.



- **A** 122 ✓
- **B** 128
- **C** 111
- **D** 112

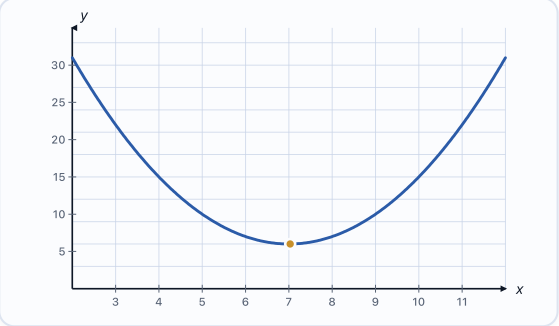
WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Conics, 3D Geometry & Probability):

1. Complete the square: $f_{933}(x) = (x - 6)^2 + 11$, so $k = 11$. **[M1]**
2. Evaluate $f_{933}(16) = (16 - 6)^2 + 11 = 100 + 11 = 111$. **[M1]**
3. Sum: $k + f_{933}(16) = 11 + 111 = 122$. **[A1]**

18 MCQ

Analyzing Conics, 3D Geometry & Probability (problem set #934), consider the function $f_{934}(x) = x^2 - 14x + 61$ illustrated on the coordinate axes. Express $f_{934}(x)$ in the completed-square form $(x - h)^2 + k$ and evaluate $k + f_{934}(9)$.



- **A** 28 ✓
- **B** 35
- **C** 16
- **D** 26

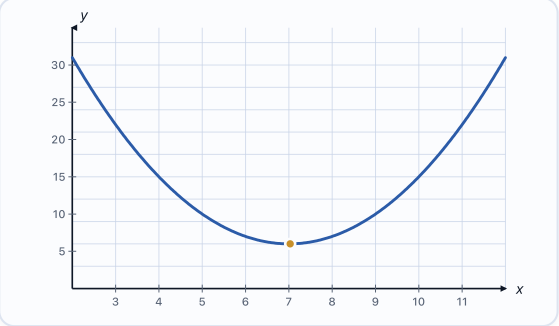
WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Conics, 3D Geometry & Probability):

1. Complete the square: $f_{934}(x) = (x - 7)^2 + 12$, so $k = 12$. **[M1]**
2. Evaluate $f_{934}(9) = (9 - 7)^2 + 12 = 4 + 12 = 16$. **[M1]**
3. Sum: $k + f_{934}(9) = 12 + 16 = 28$. **[A1]**

19 MCQ

Within Conics, 3D Geometry & Probability (problem set #935), consider the function $f_{935}(x) = x^2 - 14x + 62$ illustrated on the coordinate axes. Express $f_{935}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{935}(10)$.



A 42

B 22

C 32

D 35 ✓

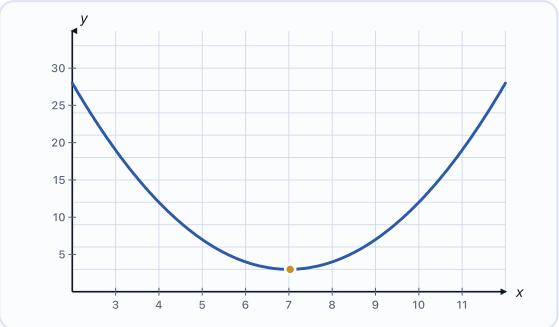
WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Conics, 3D Geometry & Probability):

1. Complete the square: $f_{935}(x) = (x-7)^2 + 13$, so $k = 13$. **[M1]**
2. Evaluate $f_{935}(10) = (10-7)^2 + 13 = 9 + 13 = 22$. **[M1]**
3. Sum: $k + f_{935}(10) = 13 + 22 = 35$. **[A1]**

20 MCQ

Considering Conics, 3D Geometry & Probability (problem set #936), consider the function $f_{936}(x) = x^2 - 14x + 52$ illustrated on the coordinate axes. Express $f_{936}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{936}(11)$.



A 29

B 19

C 18

D 22 ✓

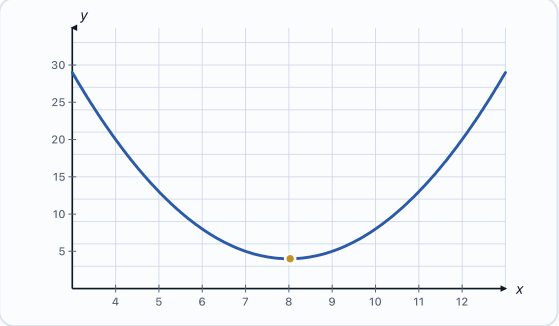
WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Conics, 3D Geometry & Probability):

1. Complete the square: $f_{936}(x) = (x-7)^2 + 3$, so $k = 3$. **[M1]**
2. Evaluate $f_{936}(11) = (11-7)^2 + 3 = 16 + 3 = 19$. **[M1]**
3. Sum: $k + f_{936}(11) = 3 + 19 = 22$. **[A1]**

21 MCQ

Examining Conics, 3D Geometry & Probability (problem set #937), consider the function $f_{937}(x) = x^2 - 16x + 68$ illustrated on the coordinate axes. Express $f_{937}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{937}(13)$.



A 41

B 29

C 28

D 33 ✓

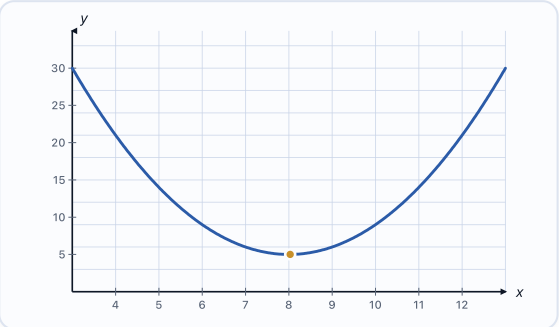
WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Conics, 3D Geometry & Probability):

1. Complete the square: $f_{937}(x) = (x-8)^2 + 4$, so $k = 4$. **[M1]**
2. Evaluate $f_{937}(13) = (13-8)^2 + 4 = 25 + 4 = 29$. **[M1]**
3. Sum: $k + f_{937}(13) = 4 + 29 = 33$. **[A1]**

22 MCQ

For the configuration in Conics, 3D Geometry & Probability (problem set #938), consider the function $f_{938}(x) = x^2 - 16x + 69$ illustrated on the coordinate axes. Express $f_{938}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{938}(14)$.



A 40

B 46 ✓

C 54

D 41

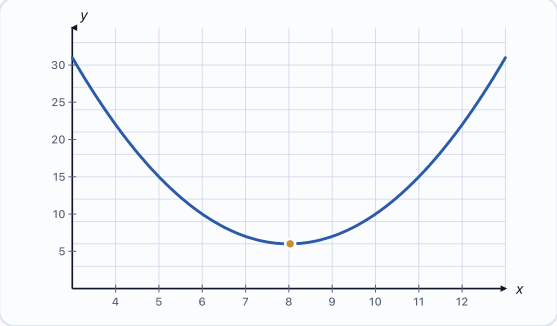
WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Conics, 3D Geometry & Probability):

1. Complete the square: $f_{938}(x) = (x-8)^2 + 5$, so $k = 5$. **[M1]**
2. Evaluate $f_{938}(14) = (14-8)^2 + 5 = 36 + 5 = 41$. **[M1]**
3. Sum: $k + f_{938}(14) = 5 + 41 = 46$. **[A1]**

23 MCQ

Analyzing Conics, 3D Geometry & Probability (problem set #939), consider the function $f_{939}(x) = x^2 - 16x + 70$ illustrated on the coordinate axes. Express $f_{939}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{939}(15)$.



- **A** 61 ✓
- **B** 54
- **C** 69
- **D** 55

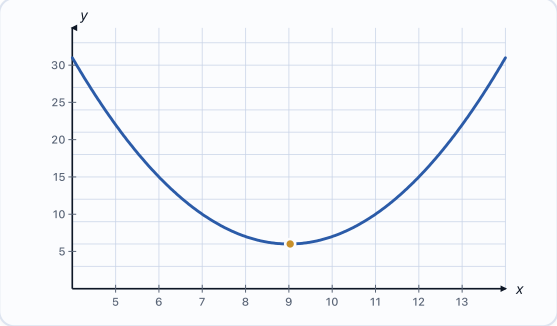
WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Conics, 3D Geometry & Probability):

1. Complete the square: $f_{939}(x) = (x-8)^2 + 6$, so $k = 6$. **[M1]**
2. Evaluate $f_{939}(15) = (15-8)^2 + 6 = 49 + 6 = 55$. **[M1]**
3. Sum: $k + f_{939}(15) = 6 + 55 = 61$. **[A1]**

24 MCQ

Within Conics, 3D Geometry & Probability (problem set #940), consider the function $f_{940}(x) = x^2 - 18x + 88$ illustrated on the coordinate axes. Express $f_{940}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{940}(17)$.



- **A** 71
- **B** 70
- **C** 87
- **D** 78 ✓

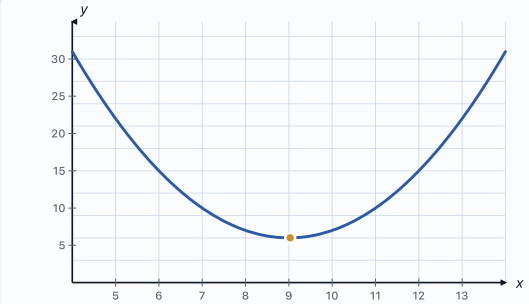
WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Conics, 3D Geometry & Probability):

1. Complete the square: $f_{940}(x) = (x-9)^2 + 7$, so $k = 7$. **[M1]**
2. Evaluate $f_{940}(17) = (17-9)^2 + 7 = 64 + 7 = 71$. **[M1]**
3. Sum: $k + f_{940}(17) = 7 + 71 = 78$. **[A1]**

25 MCQ

Considering Conics, 3D Geometry & Probability (problem set #941), consider the function $f_{941}(x) = x^2 - 18x + 89$ illustrated on the coordinate axes. Express $f_{941}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{941}(18)$.



A 106

B 89

C 88

D 97 ✓

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Conics, 3D Geometry & Probability):

1. Complete the square: $f_{941}(x) = (x-9)^2 + 8$, so $k = 8$. **[M1]**
2. Evaluate $f_{941}(18) = (18-9)^2 + 8 = 81 + 8 = 89$. **[M1]**
3. Sum: $k + f_{941}(18) = 8 + 89 = 97$. **[A1]**